Project Management Plan

for

THEIA: Indoor Navigation App for the Blind and Visually Impaired

Version 1.0 draft 1

Prepared by Ubuntu Bros.

Reviewed by

Cpt S 484, Washington State University, Fall 2026

9/19/2026

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Revision History

Name	Date	Reason for Changes	Version
All	9/19/2026	initial draft	1.0 draft 1

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<Note: This template contains primarily guidance text, shown in italics. When creating a project management plan from this template, replace the guidance text with your own specific information for the project and change the Normal style for the document to be normal font, not italic. If a section of this template is not applicable to your project, leave the section heading in the plan and briefly state why it does not apply. Feel free to tailor this comprehensive template to best meet the needs of your organization's projects.>

1. Overview

This plan describes how Ubuntu Bros will carry out Phase I of the Cpt S 484 team project. In Phase I, the team will study the informal Preliminary Definition of THEIA, a smartphone app designed to help aid blind and visually impaired (BVI) people navigate indoors. Using the information gathered from the Preliminary Definition, the team will the produce a WRS document, AS-IS/TO-BE scenarios, a recorded presentation, a mock-up prototype, user manual, and an AI Usage and Teamwork report. The schedule is fixed to future course due dates and there is no budget; therefore the plan will be to divide the work fairly, keeping the work load coherent and streamlined to be completed at least a day before the due date to give the team time to edit and reread. The full planned schedule is available at Section 3.4.

1.1. Project Purpose, Objectives, and Success Criteria

Purpose. The purpose of the project is to practice software requirements engineering on a realistic problem at scale, while avoiding the dangers of working on a real project. This creates valuable experience through a simulated and controlled environment, allowing the team to learn the software requirements processes before working on real world projects.

Scope. This plan covers Phase I only. Phase I introduces requirements models, a requirements specification, and a mock-up only prototype. It does not include a working implementation, design documents, or any other requirements an average project would require such as testing.

Objectives. The team has set the following measurable objectives for Phase I:

- **OBJ-1:** Identify and document issues in all three sections of Preliminary Definition, and give a resolution and design rationale for every issue.
- **OBJ-2:** Produce an Improved Understanding in the WRS document that breaks down the document into Goals, Objectives, and Requirement specifications; each should have a flow of Requirement Specification traces to at least one Objective, every Objective to a Goal, and every Goal to a Problem.
- **OBJ-3:** Present four AS-IS/TO-BE scenario pairs and an analysis of the Top Priority TO-BE scenario in a recorded presentation.
- **OBJ-4:** Build a mock-up prototype that is designed for BVI use and a user manual that shows users common TO-BE scenarios
- **OBJ-5:** Submit a preliminary Project Plan for feedback, update with feedback during Phase I, and submit a final version at the end of Phase I.
- **OBJ-6:** Submit an AI Usage and Teamwork Report with evidence of any Gen AI use and a description of how the output has been reviewed

Success Criteria. The project will be considered successful if all six deliverables are submitted on or before their due dates; every document section has been reviewed by at least one team member other than its author; terminology and identifiers are consistent across all documents; instructor feedback on the preliminary Project Plan is incorporated into the final version; task distribution is reflected in each member's contributions.

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Relationship to other work. The WRS document, user manual, and Project Plan will be further refined and finished in Phase II.

- **Shared resources**: team git repository, course templates, and samples on Canvas
- **Feature dependencies:**
    1. The Improved Understanding depends on the Issues and their design rationale.
    2. The TO-BE scenarios depend on the Requirements Specifications.
    3. The prototype and user manual depend on the TO-BE scenarios.
- **Schedule Dependencies:** instructor feedback on preliminary plan is needed before finalized plan and continuing of rest of project

1.2. Project Deliverables

<List the major items to be delivered to the customers, subcontractors, integrators, or other parties. As appropriate, list the deliverables, interim and final delivery dates, and delivery method. A table like the one below is a good way to show this information.>

Deliverable	Delivery Date	Delivery Method	Comments
Preliminary Project Plan	9/20/2026	Canvas Submission/GitHub Repository	
WRS Document: Issues (III.2.1)	TBA	Canvas Submission/GitHub Repository	
WRS Document: Improved Understanding (III.2.2)	TBA	Canvas Submission/GitHub Repository	
AS-IS/TO-BE Slides and Presentation	TBA	Canvas Submission/GitHub Repository	
Project Prototype/User Manual	TBA	Canvas Submission/GitHub Repository	
Final Project Plan	10/18/2026	Canvas Submission/GitHub Repository	
AI Usage and Team Report	TBA	Canvas Submission/GitHub Repository	

1.3. Assumptions, Dependencies, and Constraints

ID	Description
AS-1	The Preliminary Definition, templates, and samples posted on Canvas remain during Phase I.
AS-2	Each team member can commit approximately 2-3 hours per week to the project, including team meetings.

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ID	Description
AS-3	The team may not be able to interview actual BVI users. If not, the team will use personas and published accessibility guidance to represent users' needs.
AS-4	A mock up prototype is sufficient for Phase I.
DE-1	Due dates for all deliverables are set by the Course Schedule.
DE-2	The final Project Plan depends on instructor feedback on the preliminary submission.
DE-3	The AS-IS/TO-BE scenarios depend on the Lesson 6 material and the Theia-AS-IS_and_TO-BE_v2.0.pptx template; the WRS depends on the WRS template, incomplete sample, and complete example document.
DE-4	The team depends on third-party services (GitHub, YouTube, recording software, Gen AI tools) being available.
CO-1	Schedule: deadlines are fixed by the course and cannot be moved by the team.
CO-2	Staff: the team size is fixed at 5 members for the semester.
CO-3	Cost: there is no budget; only free or university-licensed tools will be used.
CO-4	The recorded presentation must be no more than 10 to 15 minutes long.
CO-5	Only Phase I deliverables are in scope; hypothetical deliverables such as test plans or design documents are excluded.
CO-6	Any Gen AI use must be disclosed with evidence, and the team remains fully responsible for the correctness of all submissions.

Five Project Dimensions Table.

Dimension	Classification	Rationale
Schedule	Constraint	Due dates are fixed by the Course Schedule (CO-1).
Staff	Constraint	Team membership is fixed (CO-2).
Cost	Constraint	No budget is available (CO-3).
Quality	Success driver	Deliverables are graded on completeness, consistency, traceability, and coherence.
Features	Degree of freedom	The team decides which missing requirements to add and how to resolve each issue, guided by its design rationale.

1.4. References

- Cpt S 484 Team Project Specification, Phase I: Requirements Elicitation, Initial Understanding, v2.0 (adopted from Dr. Lawrence Chung's course at UTD). Canvas.
- Cpt S 484 Project THEIA: Preliminary Definition. Canvas.
- WRS document template, incomplete sample, and complete example document for another project. Canvas.
- Theia-AS-IS_and_TO-BE_v2.0.pptx presentation template. Canvas.

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- Cpt S 484 Lesson 6: AS-IS/TO-BE scenarios. Canvas.
- Cpt S 484 Course Schedule. Canvas.
- Project Management Plan template (ProjectPlanTemplate_updated.docx) and sample project plans. Canvas.
- F. P. Brooks, "No Silver Bullet: Essence and Accidents of Software Engineering," 1987.
- Team Git Repository:

1.5. Definitions and Acronyms

Term / Acronym	Definition
THEIA	The generic name of the app (after the Greek goddess of vision): a smartphone app that helps blind and visually impaired people navigate indoors.
BVI	Blind and Visually Impaired: the primary users of THEIA.
WRS	World, Requirements, Specification. The main Phase I document, which describes the domain (World), the requirements, and the system specification, and the traceability between them.
AS-IS scenario	A description of how a task is performed today, without the proposed system, including its problems.
TO-BE scenario	A description of how the same task will be performed with the proposed system.
FR / NFR	Functional Requirement / Non-Functional Requirement.
RS	Requirement Specification: the most detailed level of the WRS hierarchy, describing specific features of the app.
Traceability	The documented links between Problems, Goals, Objectives, and Requirement Specifications.
Mock-up	A low-fidelity prototype (drawings or a GUI demonstration) showing how users will interact with the app.
Gen AI	Generative artificial intelligence tools, such as large language model chat assistants.

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2. Project Organization

The team's main external entities are the course instructor and the TAs, who act as the evaluator for each deliverable; the Team Liaison will handle communication with them. Deliverables will be submitted through Canvas and GitHub.

2.1. Process Model

The team will follow an iterative requirement engineering process. The phases below overlap, with some phases requiring parts from previous phases to build off and polish.

Phase	Key Tasks	Outputs
1. Planning and team setup	Set up git repository and communication channel; read the specification, Preliminary Definition, and templates; assign tasks.	Preliminary Project Plan
2. Elicitation and analysis	Analyze each statement of the Preliminary Definition; classify issues; propose resolutions with design rationale.	WRS: Issues
3. Specification	Identify Problems; derive Goals, functional and non-functional Objectives, and Requirement Specifications; build the traceability matrix; clarify World and System descriptions.	WRS: Improved Understanding
4. Validation	Write AS-IS/TO-BE scenario pairs and Top Priority analysis; build the mock-up; write the user manual; peer-review all documents for consistency and traceability.	Slides, prototype, user manual
5. Presentation and submission	Record and upload the presentation; compile the AI Usage and Teamwork Report; finalize the Project Plan; submit.	Video link, AI report, final Project Plan

2.2. Organizational Structure

The team is a small, self-organizing group of 5 members. The Team Liaison will ultimately be in charge of coordinating the work to complete the objectives the day before they are due; every other member is responsible for splitting up the work evenly and completing the objectives. Each deliverable will have one owner and one reviewer.

- **Meetings:** team meetings on a need basis. Minutes, decisions, and future actions are recorded in the GitHub Repository
- **Communication:** communication will occur through outlook. Members are expected to reply within 24 hours.
- **Work tracking:** tasks are tracked on GitHub, as well as milestones to be communicated through team through outlook
- **Decision making:** decisions are made through total consensus.

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2.3. Roles and Responsibilities

Role	Member	Responsibilities
Team Liaison / Project Manager	Jacob Atwood	Single point of contact with the instructor; schedules and runs meetings; maintains this Project Plan and the task board; tracks risks; performs the cross-section consistency review of the Issues; submits deliverables on Canvas.
Requirements Analyst (Issues Lead)	Eniola	Leads the Issues part of the WRS; merges issues from all members into one consistent format; checks that every statement in the Preliminary Definition has been analyzed.
Requirements Analyst (Specification and Traceability Lead)	Eniola	Leads the Improved Understanding; owns the Problem, Goal, Objective, and RS identifiers and the traceability matrix.
UX and Accessibility Designer	David Acosta Leon	Leads the mock-up prototype and the preliminary user manual; ensures the design supports non-visual interaction.
Presentation Lead	Jaydon Devictoria	Leads the AS-IS/TO-BE slides and Top Priority analysis; coordinates recording, editing, timing (10 to 15 minutes)
Configuration Manager and Document Editor	Travis Takushi	Maintains the git repository structure and naming conventions; performs the final editing pass so that all documents read as one coherent submission.
User Proxy / Subject Matter Expert	All members	Contact potential users where possible; otherwise role-play BVI users and review requirements from their point of view.

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3. Managerial Process Plans

This section describes how the team will manage its work, risks, and schedule during Phase I.

3.1. Management objectives and priorities

- **Meet every deadline.** Each deliverable will have a due date of 1 day before the Canvas due date, reserved for review
- **Maximize quality of WRS document.** Document must read as one team's work, as well as holds the most weight of the grade
- **Keep workload balanced and visible.** Work is assigned through discussion, and will be reviewed after every milestone to rebalance

3.2. Assumptions, dependencies, and constraints

- Members are expected to respond to direct outlook messages within 24 hours
- Individual contributions are recorded through git commits, the task board, and meeting minutes; these records are also used for the Teamwork part of the AI Usage and Teamwork Report.
- If a member is unresponsive for more than 3 days, the Team Liaison reassigns their tasks; if the problem continues, the Liaison informs the instructor.
- All deliverables are stored in the team git repository before they are submitted on Canvas.

3.3. Risk management

ID	Risk	Prob.	Impact	Mitigation / Contingency	Owner
R-1	A member becomes unavailable (illness, workload, dropping the course).	M	H	Every deliverable has a reviewer who knows its content; the Liaison reassigns tasks within 24 hours.	
R-2	Sections written by different members are inconsistent in terminology, identifiers, or style.	H	H	Shared glossary and ID scheme agreed in week 1; final editing pass by the Document Editor.	
R-3	Gen AI output contains errors or hallucinations that reach a submission.	M	H	Every AI-assisted passage is checked against the specification and Preliminary Definition by a second member; AI use is logged.	

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ID	Risk	Prob.	Impact	Mitigation / Contingency	Owner
R-4	The team misinterprets the scope or format of a deliverable.	M	H	Use the preliminary plan feedback; compare drafts with the samples; ask the instructor early through the Liaison.	
R-5	Traceability gaps (requirements with no objective, or objectives with no requirement).	M	H	Maintain the traceability matrix as the WRS is written, not at the end; check it before each review.	
R-6	Git merge conflicts or lost work on binary files (Word, PowerPoint).	M	M	One editor per file at a time; commit often; keep dated backups of binary files.	
R-7	Scope creep (adding out-of-scope deliverables or too many features).	L	M	Check every new item against the specification and constraint CO-5.	

3.4. Schedule and Milestones

Wk	Dates	Activities	Owner(s)	Milestone
1	9/20/2026	Set up git repository; read the specification, Preliminary Definition, and templates; agree on glossary and ID scheme; draft the Project Plan.	all	Preliminary Project Plan submitted
2	TBA	Analyze each section of the Preliminary Definition; attempt to contact potential users; create personas; start the issue log.	Travis Takushi Eniola: preliminary definition	Issue log started
3	TBA	Classify and resolve issues with design rationale; cross-review sections; incorporate instructor feedback on the plan.	all	Issues draft complete
4	TBA	Write Problems, Goals, Objectives, and Requirement Specifications; build the traceability matrix; draft four AS-IS/TO-BE pairs.	Jaydon: scenario pairs only	Improved Understanding draft
5	TBA	Build the mock-up prototype; write the user manual from the TO-BE scenarios; prepare slides and Top Priority analysis.	David: prototype and manual; Jaydon: slides	Full WRS draft; slides draft

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Wk	Dates	Activities	Owner(s)	Milestone
6	TBA	Full-team review and editing pass; rehearse and record the presentation; compile the AI Usage and Teamwork Report; finalize the Project Plan.	All	All Phase I deliverables submitted

4. Technical Process Plans

Phase I is for requirements engineering, focused on analyzing and validating requirements rather than writing code. The only technical implementation will be the mock-up prototype.

4.1. Methods, tools, and techniques

- **Software Tools:** GitHub: version control and tasks, Microsoft Word: drafting, PowerPoint: presentations for AS-IS/TO-BE, Zoom: recording/editing, Canvas: submitting project deliverables
- **Requirements methods:** document analysis of Preliminary Definition, team brainstorming for models, Issues are classified as incompleteness, inconsistency, ambiguity, redundancy, or unsoundness, and each resolution records the chosen option, the alternatives considered, and the reason for the choice
- **Specification conventions:** the WRS template structure (World, Requirements, Specification); a Problem, Goal, Objective, and Requirement Specification hierarchy with unique IDs (for example P-1, G-1, FO-1, NFO-1, FR-1, NFR-1); a traceability matrix linking all levels; AS-IS/TO-BE scenarios following Lesson 6; and a shared glossary for consistent terminology.
- **Prototype approach:** the mock-up treats audio feedback, haptic (vibration) cues, voice commands, and screen-reader compatibility as the primary interface, with large, high-contrast visual elements as a secondary aid for low-vision users.

4.2. Software documentation

Document	Template or Standard	Created By	Reviewed By	Target Date	Distribution
Project Plan (preliminary and final)	ProjectPlanTemplate_updated.docx; course samples	All(Preliminary); All(Final)	All	9/20/2026	Canvas; git

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Document	Template or Standard	Created By	Reviewed By	Target Date	Distribution
WRS Document (Issues and Improved Understanding)	WRS template, sample, and example on Canvas	Travis Takushi (Issues), Eniola (Improved Understanding)	Travis Takushi	TBA	Canvas; git
AS-IS/TO-BE slides and recorded presentation	Theia-AS-IS_and_TO-BE_v2.0.pptx; Lesson 6	Jaydon Devictoria; all members	All	10/17/2026	Slides in git; video on YouTube (unlisted), link on Canvas
Mock-up prototype	Team conventions; platform accessibility guidelines	David Acosta Leon	Jaydon Devictoria	10/17/2026	Canvas; GitHub
Preliminary user manual	WRS template section (or standalone)	David Acosta Leon	Jaydon Devictoria	10/17/2026	Canvas; GitHub
AI Usage and Teamwork Report	No template; includes evidence	Travis Takushi		TBA	Canvas; GitHub
Meeting minutes (internal)	Team minutes format	Rotating		Request Basis	GitHub only

4.3. David and Jaydon Phase I Work Plan

This is our plan for the prototype, user manual and scenario presentation. David Acosta Leon will take the UX and Accessibility Designer role. Jaydon Devictoria will take the Presentation Lead role. We will work from the team's WRS so the examples, app screens and instructions describe the same system. These assignments cover our contributions. The remaining lead roles are still open for the other team members to fill in.

4.3.1 David: prototype and preliminary user manual

David will build a simple mock-up in PowerPoint and write the preliminary user manual in Word. The mock-up will show how a person starts a trip, confirms a destination, follows directions, responds to an obstacle and asks for help. It will also show how the user changes the volume and frequency of instructions. This is a demonstration of the interaction. It does not need working indoor positioning, sensors or emergency calls for Phase I.

The main idea is that a user should not have to see the screen to understand the next step. Each important screen will include the spoken prompt, the action the user can take and the feedback they receive. We will show options to repeat an instruction, go back and cancel. Large text and

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clear contrast will support people with low vision. Audio and vibration cues will be described in the mock-up without claiming that accessibility has already been tested with BVI users.

The manual will walk through the same four TO-BE scenarios used in the slides. It will cover getting started, choosing a destination, following directions, handling an obstacle, asking for help and changing settings. Each walkthrough will explain what the user does and what THEIA says or does next. If a destination is unclear or the app loses the user's location, the example will pause navigation and explain the problem instead of giving another direction that could be wrong.

David's draft is ready for review when every scenario has a matching flow and the manual follows it without missing steps. Jaydon will walk through the mock-up using the manual and flag any step that depends only on sight. David will correct the gaps and keep the final mock-up and manual in the team repository.

4.3.2 Jaydon: scenarios and Phase I presentation

Jaydon will use Theia-AS-IS_and_TO-BE_v2.0.pptx to prepare four scenario pairs. Each pair will show the user, the problem without THEIA and how the proposed app helps. The starting set is below. We will refine it with the team and link each TO-BE scenario to the matching WRS requirement IDs once those IDs are agreed.

1. Stevie gets to his next classroom. In the given AS-IS example Stevie is unsure when to turn, turns too early and hits the wall. The TO-BE example will show THEIA confirming his destination and giving a clear instruction at the turn. It will also show how he can repeat or stop the guidance.

2. A user comes across a blocked hallway. Without a warning the user may continue toward an unexpected obstacle. The TO-BE example will show an obstacle warning and a choice to stop or follow another route. The mock-up will state the assumed obstacle information so it does not suggest that a phone can reliably detect every hazard.

3. A visitor gets lost and needs help. In the AS-IS example the visitor cannot describe their location well enough to get directions. The TO-BE example will show THEIA explaining that location is uncertain, pausing the route and letting the user confirm a request to a chosen support contact. It will show what happens if the contact cannot be reached. No real call or message is needed for the demonstration.

4. A user sets up instructions with a caretaker. In the AS-IS example the instructions are too quiet or too frequent and the user struggles with the settings. The TO-BE example will show spoken setting choices, a preview and confirmation before saving. The user keeps control of the settings even when a caretaker helps.

Our starting choice for the top priority scenario is Stevie's classroom trip because it combines a common task with a direct safety concern. Jaydon will include at least one analysis slide using the Lesson 6 questions: who starts or stops the trip, what kinds of routes are supported, when a direction should be given, how the user repeats it and what if the destination is unclear, a turn is missed or location information is lost. The team's answers will be used to improve the scenario and the related WRS requirements.

Jaydon will coordinate the recording after the team reviews the slides. The target is 10 to 15 minutes, covering all four pairs and the top priority analysis. David will check that the slides match the prototype and manual. The final package will include the editable slides in the repository and an unlisted video link for the team's Canvas submission.

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4.3.3 Our schedule and review points

The dates below are our internal targets for 2026. The preliminary plan is due September 20 and the Phase I package is due October 18 at 11:59 pm Pacific. We will finish our final files by October 17 so the team has time to assemble the submission.

Dates	Work and owner	Review or milestone
Sept. 19 to 20	David and Jaydon finish their roles and work plan for the preliminary submission.	Read our plan together before the team submits it.
Sept. 21 to 27	Jaydon drafts four scenario pairs. David outlines the interaction flows and manual sections. Eniola works on the pleminary definition for the project and review	Compare the drafts with the Preliminary Definition on Sept. 28.
Sept. 28 to Oct. 4	David creates the first mock-up. Jaydon develops the top priority analysis and checks the scenario links with the WRS authors.	Walk through the first mock-up on Oct. 5. Record missing steps and open questions.
Oct. 5 to 11	David completes the prototype and manual drafts. Jaydon completes the slide draft with all four pairs and the analysis.	Full drafts ready Oct. 11. Cross-review Oct. 12 to 14.
Oct. 12 to 14	David fixes the prototype and manual. Jaydon fixes the scenarios and slides using team feedback.	Check terminology, WRS links, alternate flows and non-visual instructions.
Oct. 15 to 17	Jaydon coordinates rehearsal and recording on Oct. 15 to 16. Both check their final files and AI-use evidence.	Check video sound, length and link access. Final files ready Oct. 17 for the Oct. 18 team submission.

4.3.4 Coordination and completion checks

We will use the Requirements Review Checklist during our cross-review, especially completeness, clear wording, error conditions and traceability. If the WRS changes, we will update the affected scenario, mock-up flow and manual steps together. Unresolved decisions will be marked TBD and taken to the relevant WRS lead. If the prototype takes longer than planned, David will use annotated screens so the required flows are still covered. If recording has a problem, Jaydon will use the saved slides and allow time for another recording before the internal deadline. Eniola will check that team follows every statement in the Preliminary Definition has been analyzed and also goal objective.

For the AI Usage and Teamwork Report, Codex assisted with drafting and organizing these additions to the preliminary plan. David and Jaydon will keep the prompt and output as evidence and review the text themselves before submission. They will record the corrections they make and add any later AI use in their own work. This statement covers these additions only and does not describe other teammates' use of AI.

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